

求定积分

$$\int_0^\pi \frac{\pi + \cos x}{x^2 - \pi x + 2026} \mathrm{d}x$$

令 $x - \frac{\pi}{2} = t$
得到

$$\begin{aligned} & \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\pi - \sin t}{t^2 - \frac{\pi^2}{4} + 2026} \mathrm{d}t \\ &= 2\pi \int_0^{\frac{\pi}{2}} \frac{\mathrm{d}t}{t^2 - \frac{\pi^2}{4} + 2026} \\ &= \frac{2\pi}{\sqrt{2026 - \frac{\pi^2}{4}}} \left(\arctan \frac{\pi}{2\sqrt{2026 - \frac{\pi^2}{4}}} \right) \end{aligned}$$